

**Q5 Motion under gravity A
level only**

Name: _____

Class: _____

Date: _____

Time: **287 minutes**

Marks: **237 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

A football is placed on a horizontal surface. It is then kicked, so that it has an initial velocity of 12 m s^{-1} at an angle of 40° above the horizontal.

- (a) State two modelling assumptions that it would be appropriate to make when considering the motion of the football. (2)
- (b) (i) Find the time that it takes for the ball to reach its maximum height. (4)
- (ii) Hence show that the maximum height of the ball is 3.04 metres, correct to three significant figures. (3)
- (c) After the ball has reached its maximum height, it hits the bar of a goal at a height of 2.44 metres. Find the horizontal distance of the goal from the point where the ball was kicked. (7)

(Total 16 marks)

Q2.

An arrow is fired horizontally with a speed of 30 m s^{-1} from a height of 2 m above ground level. Model the arrow as a particle which does not experience air resistance.

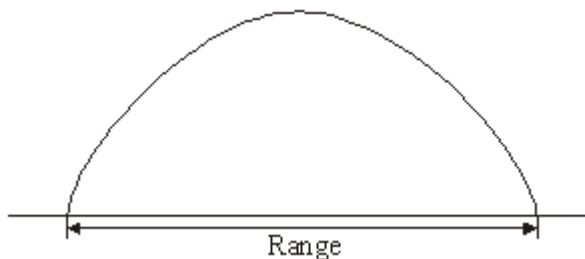
- (a) Show that the time taken for the arrow to hit the ground is 0.639 seconds, correct to three significant figures. (4)
- (b) Find the horizontal distance travelled by the arrow. (2)
- (c) Describe what happens, during the flight of the arrow, to the magnitude of:
 - (i) the horizontal component of the velocity;
 - (ii) the vertical component of the velocity.

(2)
(Total 8 marks)

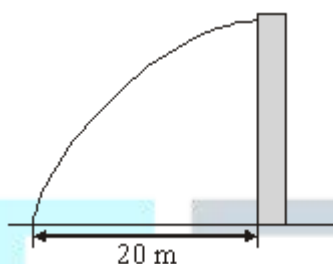
Q3.

A ball is hit from horizontal ground with velocity $(10\mathbf{i} + 24.5\mathbf{j}) \text{ m s}^{-1}$ where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertically upwards respectively.

- (a) State two assumptions that you should make about the ball in order to make predictions about its motion. (2)
- (b) The path of the ball is shown in the diagram.



- (i) Show that the time of flight of the ball is 5 seconds. (4)
- (ii) Find the range of the ball. (2)
- (c) In fact the ball hits a vertical wall that is 20 metres from the initial position of the ball.



- Find the height of the ball when it hits the wall. (4)
- (d) If a heavier ball were projected in the same way, would your answers to part (b) of this question change? Explain why. (2)
- (Total 14 marks)

Q4.

A golf ball is driven with speed 40 m s^{-1} at an angle α to the horizontal from a point on a horizontal golf course. The ball travels in a vertical plane.

- (a) Show that, during its flight, the horizontal and vertical displacements, x metres and y metres respectively, of the ball from the point of projection satisfy the equation

$$y = x \tan \alpha - \frac{gx^2}{3200} (1 + \tan^2 \alpha) \quad (5)$$

- (b) The golf ball just clears a tree of vertical height 4 m at a horizontal distance of 100 m from its point of projection.

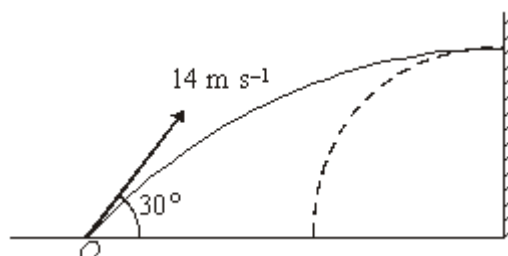
Find the two possible values of α .

- (c) State two modelling assumptions that you have made. (6)

(2)
 (Total 13 marks)

Q5.

Adam kicks his ball from a point O on a horizontal surface towards a vertical wall. He kicks the ball with speed 14 m s^{-1} at an angle of 30° to the horizontal, as shown in the diagram.



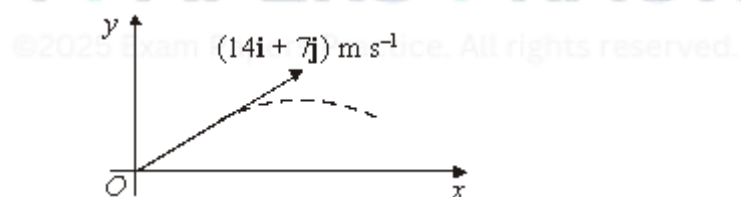
When the ball hits the wall, it is travelling **horizontally**.

- (a) Find the vertical height above O at which the ball hits the wall. (4)
- (b) The ball rebounds horizontally from the wall. Find the time that the ball takes to reach the horizontal surface. (3)

(Total 7 marks)

Q6.

Jonny is practising kicking a rugby ball. He kicks the ball from a point O on horizontal ground with velocity $14\mathbf{i} + 7\mathbf{j} \text{ m s}^{-1}$. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in the directions of axes Ox and Oy , and Oy is vertically upwards.

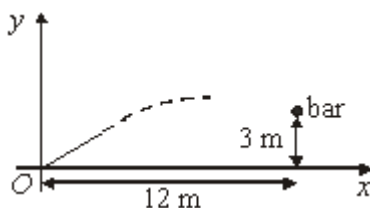


- (a) (i) Find the coordinates of the ball t seconds after it has been kicked. (3)
- (ii) Show that the equation of the path of the ball is

$$y = \frac{20x - x^2}{40}$$

- (iii) Find the horizontal range of the ball. (2)

- (b) The ball moves towards a horizontal bar which is 3 metres above the ground and at a horizontal distance of 12 metres from O , as shown in the diagram.



Determine whether the ball passes above or below the bar.

(3)

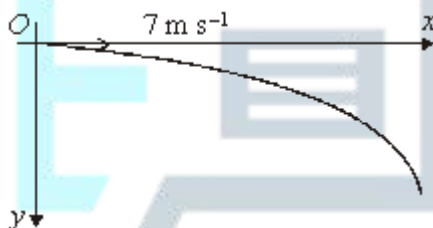
- (c) State one modelling assumption that you have used in answering this question.

(1)

(Total 12 marks)

Q7.

Elaine kicks a ball off a cliff top. She kicks the ball from a point O and it subsequently moves in a vertical plane with respect to axes which are horizontal and vertically downwards, as shown in the diagram.



The initial velocity of the ball is horizontal and of magnitude 7 m s^{-1} .

- (a) Find the coordinates of the ball t seconds after it has been kicked.

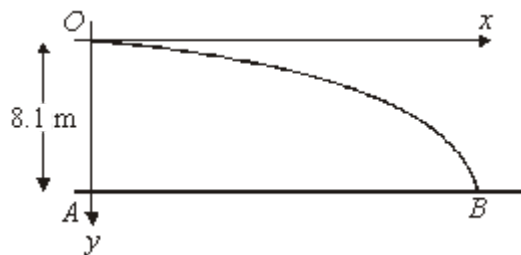
(3)

- (b) Show that the equation of the path of the ball is

$$y = \frac{x^2}{10}.$$

(2)

The ball subsequently lands on a horizontal beach. The point A on the beach is vertically below O and the ball lands at the point B on the beach, as shown in the diagram.



- (c) The distance OA is 8.1 metres.

Find the distance AB .

(2)

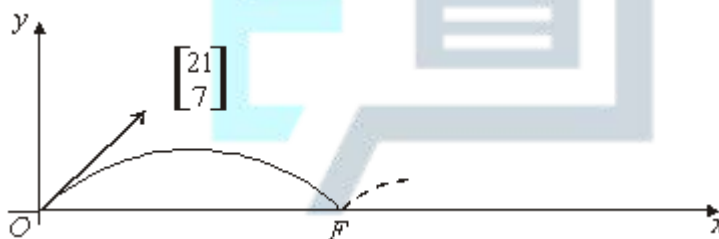
- (d) Find the speed of the ball as it reaches B .

(5)

(Total 12 marks)

Q8.

A ball is kicked from a point O on horizontal ground with velocity $\begin{bmatrix} 21 \\ 7 \end{bmatrix} \text{ m s}^{-1}$. The ball subsequently hits the ground at a point F , as shown in the diagram.



- (a) (i) Find the time between the ball being kicked and reaching F .

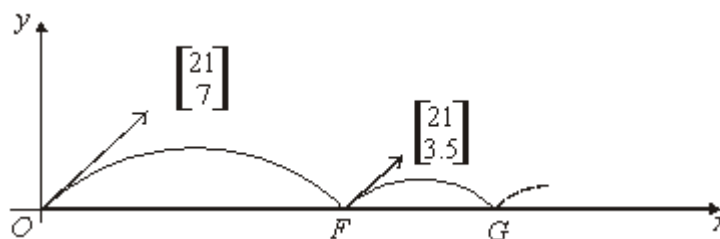
(3)

- (ii) Find the distance OF .

(2)

- (b) The ball rebounds from F with velocity $\begin{bmatrix} 21 \\ 3.5 \end{bmatrix} \text{ m s}^{-1}$. It subsequently hits the ground again at

G and then rebounds, as shown in the diagram below.



Find the distance FG .

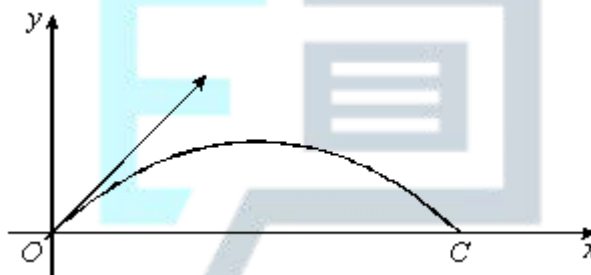
(4)

- (c) After bouncing at G , the ball rebounds with velocity $\begin{bmatrix} 21 \\ 1.75 \end{bmatrix}$ m s⁻¹, so that the vertical component of the velocity is halved again, and then subsequently bounces next at H .
Find the total distance OH .

(3)

(Total 12 marks)

Q9.



Oliver throws a ball from point O , with velocity $\begin{bmatrix} 7 \\ 7 \end{bmatrix}$ m s⁻¹. The ball is subsequently caught by

Chris at a point C which is on the same horizontal level as O , as shown in the diagram. During the motion from O to C the ball moves in a vertical plane.

- (a) Taking O as the origin, find the position vector of the ball t seconds after being thrown.

(3)

- (b) Show that the equation of the path of the ball is

$$y = x - 0.1x^2,$$

where (x, y) are the coordinates at time t of the ball relative to the origin O .

(4)

- (c) A horizontal wire, which is part of a fence, cuts the plane of motion of the ball at the point $(4, 2)$.

Determine whether the ball passes above or below the wire.

(3)

(d) Find the distance OC .

(4)

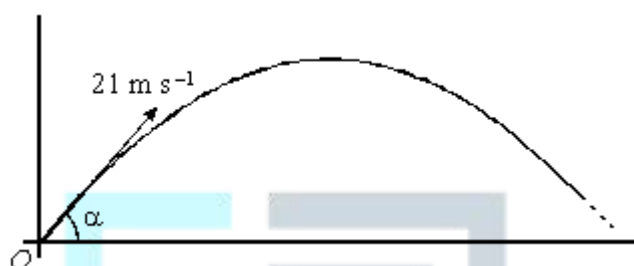
(e) Find the greatest speed of the ball during its motion from O to C .

(2)

(Total 16 marks)

Q10.

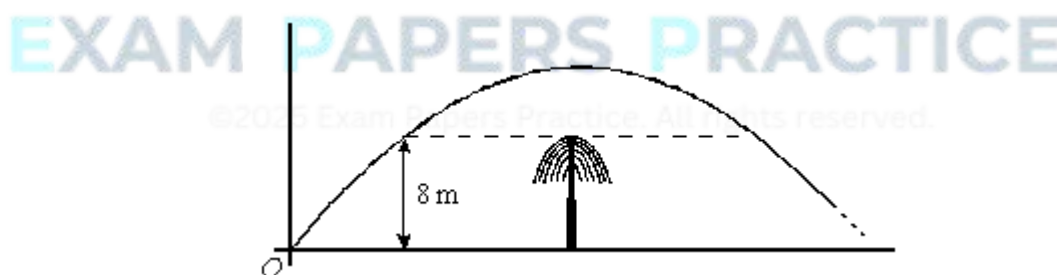
Paul throws a ball from point O with a velocity 21 m s^{-1} at an angle of α to the horizontal, where $\sin \alpha = 0.7$. The ball subsequently moves freely under gravity in a vertical plane, as shown in the diagram.



(a) Show that the time taken for the ball to reach its greatest height above O is 1.5 seconds.

(3)

(b) When the ball reaches its greatest height, it passes over a tree of vertical height 8 metres, as shown in the diagram below.



(i) Find the vertical distance between the ball and the top of the tree at this time.

(4)

(ii) Find the time between the ball leaving O and first reaching the horizontal level of the top of the tree. Give your answer to two decimal places.

(5)

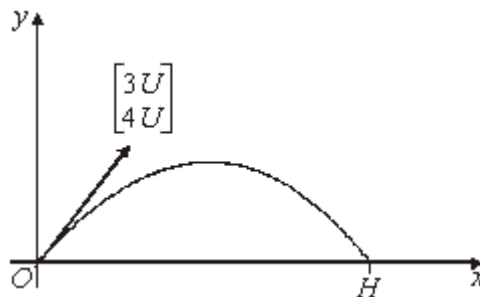
(iii) Find the length of time for which the ball is above the horizontal level of the top of the tree.

(2)

(Total 14 marks)

Q11.

George throws a ball from a point O , with velocity $\begin{bmatrix} 3U \\ 4U \end{bmatrix}$. The ball subsequently lands at a point H , which is at the same horizontal level as O , as shown in the diagram.



- (a) Show that the time taken by the ball to travel from O to H is $\frac{8U}{g}$. (3)
- (b) Find, in terms of g and U , the distance OH . (2)
- (c) Find, in terms of U , the initial speed of the ball. (2)
- (d) Find, in terms of g and U , the two times during the flight from O to H when the ball is moving with speed $\sqrt{18}U$. (6)

(Total 13 marks)

Q12.

During a football match, a player kicks a ball from rest towards a goal, which is 12 metres from the ball. The ball is kicked with an initial velocity of 25 m s^{-1} at an angle of 17° above the horizontal.

- (a) Show that the time that it takes the ball to travel the horizontal distance of 12 metres is 0.502 seconds, correct to three significant figures. (3)
- (b) The ball hits the crossbar of the goal. Calculate the height of the crossbar. (3)
- (c) Find the angle between the velocity of the ball and the **vertical** just before the ball hits the crossbar. (5)
- (d) State two modelling assumptions that you have made. (2)

(Total 13 marks)

Q13.

A ball is kicked from ground level so that its initial velocity is 10 m s^{-1} , at an angle of 60° above the horizontal. It hits a wall, which is at a distance of 8 metres from the initial position of the ball.

- (a) Show that the ball hits the wall 1.6 seconds after it was kicked.

(3)

- (b) Find the height of the ball as it hits the wall.

(3)

- (c) Find the speed of the ball as it hits the wall.

(5)

(Total 11 marks)

Q14.

A ball is thrown so that it initially travels at 10 m s^{-1} at an angle of 70° above the horizontal.

- (a) A simple model assumes that the ball is thrown from ground level and lands at the same level. Find the time of flight and the range of the ball.

(5)

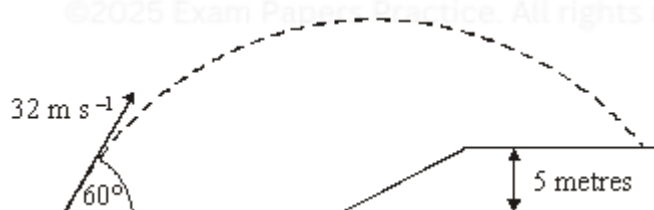
- (b) A refined model assumes that the ball is at a height of 2 metres when it is thrown, and lands at ground level. Find the range of the ball based on this assumption.

(7)

(Total 12 marks)

Q15.

A golf ball is hit so that it initially travels at 32 m s^{-1} at an angle of 60° above the horizontal. The ball lands on a horizontal surface 5 metres higher than the ground from which it was hit, as shown in the diagram below.



- (a) Show that the ball is in the air for approximately 5.47 seconds.

(5)

- (b) Calculate the horizontal distance travelled by the ball.

(2)

- (c) Find the speed of the ball when it hits the ground.

(5)

(Total 12 marks)

Q16.

A football is kicked from horizontal ground. It initially moves with speed 20 m s^{-1} at an angle of 30° above the horizontal.

- (a) (i) Show that the ball hits the ground approximately 2.04 seconds after it has been kicked.

(4)

- (ii) Hence find the range of the ball.

(2)

- (b) In fact the ball comes into contact with a player's head when it is at a height of 2 metres. Find the speed of the ball at this height.

(8)

(Total 14 marks)

Q17.

A particle is projected from a horizontal surface at a speed V and at an angle α above the horizontal.

- (a) Prove that the maximum height of the particle is $\frac{V^2 \sin^2 \alpha}{2g}$.

(6)

- (b) A ball is hit from ground level. The ball initially moves at an angle of 60° above the horizontal. The maximum height of the ball is 6 metres above the ground. Modelling the ball as a particle:

- (i) find the initial speed of the ball;

(3)

- (ii) find the range of the ball.

(4)

(Total 13 marks)

Q18.

A particle is projected from a horizontal surface and lands on the same horizontal surface. Its initial velocity is $V \text{ m s}^{-1}$ at an angle α above the horizontal.

- (a) Prove that the range of the particle is

$$\frac{V^2 \sin 2\alpha}{g}$$

(6)

- (b) Hence explain why the maximum range of the particle is $\frac{V^2}{g}$, and state the value of α for which the particle has this maximum range.

(2)

- (c) A machine for launching tennis balls fires the balls from ground level on a horizontal surface. Assume that the balls are all fired at the same speed. The maximum range of one of these balls is 35 metres.

(i) Find the speed at which the tennis balls are fired.

(2)

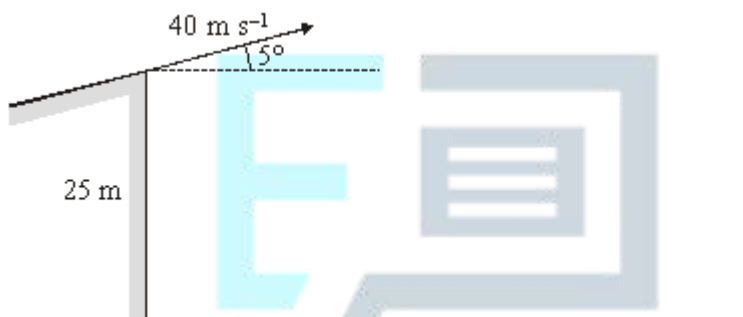
(ii) Find the range of a tennis ball fired at an angle of 30° above the horizontal, giving your answer to the nearest metre.

(2)

(Total 12 marks)

Q19.

In a film a stunt-man drives a car off the top of a vertical cliff. The top of the cliff is 25 metres above the level of the sea. When it leaves the cliff the car is travelling at 40 m s^{-1} at an angle of 5° above the horizontal. The diagram shows the cliff and the initial velocity of the car.



Model the car as a particle and assume that, while it is in the air, it moves under the influence of gravity alone.

- (a) Show that the car hits the sea approximately 2.64 seconds after it leaves the top of the cliff.

(6)

(b) Find the horizontal distance of the car from the cliff when the car hits the sea.

(2)

(c) Find the speed of the car when it hits the sea.

(5)

(Total 13 marks)